Almost sure covariance escape for scalar Gaussian ensemble Langevin discretization

Abstract

For every fixed ensemble size $N \geq 2$, fixed step size $\eta > 0$, and finite positive initial covariance, we prove almost-sure convergence of the empirical covariance to positive infinity for the untamed forward Euler ensemble Langevin scheme with a one-dimensional standard Gaussian target. We derive its exact $(N-1)$-dimensional Gaussian covariance kernel from the particle update, retaining the divisor N in the covariance. A Gaussian logarithm inequality supplies strictly positive boundary drift even when $N = 2$. Fractional inverse moments exclude eventual trapping near zero; a uniform positive escape probability and conditional-probability martingale convergence then give almost-sure escape from every positive start. The proof is analytic and self-contained apart from standard conditional-expectation convergence. Zero covariance is absorbing. No escape rate, particlewise conclusion, or novelty claim is made.

Keywords: ensemble Langevin discretization; Gaussian target; covariance escape; almost sure divergence; stochastic difference equation; inverse-moment drift

2020 Mathematics Subject Classification. Primary 60H35; Secondary 60F15, 39A50, 65C35.

1 The covariance chain and its conclusion

Fix an integer $N \geq 2$, a real number $\eta > 0$, and a deterministic initial value $0 < C_0 < \infty$. Write

$$d = N - 1, \qquad a = \frac{2\eta}{N}, \qquad e = (1, 0, \ldots, 0)^T \in \mathbb{R}^d.$$

Let $(Z_n)_{n\geq 0}$ be independent standard Gaussian vectors in $\mathbb{R}^d$. The process of interest is

$$C_{n+1} = C_n \left\| (1 + a - \eta C_n)e + \sqrt{a}\, Z_n \right\|^2. \tag{1}$$

All norms below are Euclidean. Any fixed unit vector may replace e without changing the law.

Theorem 1. *For each fixed choice of $N \geq 2$, $\eta > 0$, and deterministic $C_0 \in (0, \infty)$, the recursion (1) satisfies*

$$\mathbb{P}(0 < C_n < \infty \text{ for every finite } n) = 1, \qquad \mathbb{P}\left(\lim_{n\to\infty} C_n = +\infty\right) = 1.$$

If $C_0 = 0$, then $C_n = 0$ for every n.

The quantifiers are pointwise in the parameters and initial state. The theorem does not assert a common probability-one event for uncountably many parameter choices.

The source application is the untamed scalar-target Euler scheme of Tan and Chen [1], Section 3.1, equations (14)–(15), with empirical covariance normalized by N. That source gives a positive-probability divergence construction for sufficiently large initial covariance. Theorem 1 gives an all-positive-start, almost-sure conclusion through the argument below. No claim about priority or completeness of the related literature is intended.

2 From the particle update to the exact kernel

Let $x = (x_1, \dots, x_N)^T$ be a scalar ensemble, and define

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad P = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}^T, \qquad y = Px, \qquad c = \frac{1}{N}\|y\|^2, \tag{2}$$

where $\mathbf{1}$ is the all-ones vector. For the standard Gaussian target, whose potential is $f(t) = t^2/2$, the literal particle update in source equation (14) is

$$x_i^+ = x_i - \eta c x_i + \frac{2\eta}{N}(x_i - \bar{x}) + \sqrt{2\eta c}\,\xi_i, \qquad 1 \le i \le N. \tag{3}$$

Conditionally on the past, $\xi = (\xi_1, \dots, \xi_N)^T$ is standard Gaussian and independent of the past. This also follows from the rectangular square root in source equation (13): for $c > 0$, each independent particle-noise vector is projected onto a past-measurable vector of squared norm c. Dividing that scalar projection by $\sqrt{c}$ gives a standard Gaussian conditional law, independent of the past. Independent particle-noise vectors give independent scalar projections.

Multiplying (3) by P, and using $P\mathbf{1} = 0$ and $Py = y$, gives the centered identity in source equation (15):

$$y^+ = \left(1 - \eta c + \frac{2\eta}{N}\right) y + \sqrt{2\eta c}\, P\xi. \tag{4}$$

Choose a fixed $N \times d$ matrix Q with orthonormal columns spanning $\mathbf{1}^\perp$. Thus $QQ^T = P$ and $Q^TQ = I_d$. Set $v = Q^Ty$ and $z = Q^T\xi$. Then $\|v\|^2 = Nc$, and z is standard Gaussian in $\mathbb{R}^d$, independent of the past. For $c > 0$, put $u = v/\sqrt{Nc}$. Taking squared norms in these coordinates yields

$$c^+ = \frac{1}{N}\|Q^Ty^+\|^2 = c\big\|(1 + a - \eta c)u + \sqrt{a}\, z\big\|^2. \tag{5}$$

The dimension is $N - 1$, and the noise factor inside the norm is $\sqrt{2\eta/N}$. Both depend on the precise normalization in (2).

For completeness, a measurable rotation turns (5) into the pathwise recursion (1). For a unit vector u, let

$$O(u) = \begin{cases} I_d - 2\dfrac{(e-u)(e-u)^T}{\|e-u\|^2}, & u \ne e, \\ I_d, & u = e. \end{cases}$$

This is a measurable orthogonal matrix with $O(u)e = u$. Define $Z = O(u)^T z$ at each step. Its conditional distribution given the entire past is standard Gaussian; hence it is independent of that past. Iterated conditioning shows that the transformed noises at successive steps are iid, not merely marginally Gaussian. Norm invariance in (5) gives (1). When $d = 1$, this construction is simply a measurable sign change.

Each positive finite state has a nondegenerate Gaussian transition on the centered subspace. It therefore reaches zero with probability zero. Conditional induction and a countable union justify the rotation at every finite time almost surely; it may be defined arbitrarily on exceptional paths. If $c = 0$, the centered drift and covariance noise both vanish. Thus zero is absorbing.

Corollary 2.1. *For any fixed deterministic finite scalar ensemble with $N \geq 2$ and positive covariance* (2)*, the fixed-step update* (3)*, with any $\eta > 0$, satisfies $C_n \to +\infty$ almost surely.*

Its proof is the kernel derivation above and Theorem 1. Equal positive initial covariance gives equal covariance path law, regardless of the ensemble mean or centered direction.

Source equation (16) is an event-conditioned lower inequality on covariance, used with a specified small-noise event. It is not an unconditional transition equality. The exact kernel here comes directly from (3)–(4), and no correction to that source inequality is needed.

3 The Gaussian boundary logarithm

Write

$$W(c, z) = \left\|(1 + a - \eta c)e + \sqrt{a}\, z\right\|^2. \tag{6}$$

The essential analytic ingredient is a strictly positive logarithmic mean at $c = 0$. Its one-dimensional case requires care because a shifted Gaussian can approach zero.

Lemma 1. *If Z is a standard real Gaussian, then for every $a > 0$,*

$$G(a) := \mathbb{E} \log |1 + a + \sqrt{a}\, Z| > 0. \tag{7}$$

Consequently, for a standard Gaussian $Z \in \mathbb{R}^d$ and every $d \geq 1$,

$$L := \mathbb{E} \log W(0, Z) \geq 2G(a) > 0. \tag{8}$$

Proof. Let ϕ be the standard real Gaussian density, and put

$$J(x) = \mathbb{E} \log |x + Z| = \int_{\mathbb{R}} \log |t| \phi(t - x)\, dt.$$

The logarithmic singularity is locally integrable, and Gaussian tails control infinity. Differentiating the density under the integral is valid to every order: on each compact set of shifts, all relevant density derivatives times $|\log|t||$ have a common integrable envelope. Thus J is smooth. Set $H = J'$. Symmetry gives $H(0) = 0$.

Direct differentiation of the density gives

$$J''(x) + xJ'(x) = \int_{\mathbb{R}} \log|t|\,[t(t-x)-1]\phi(t-x)\,dt = 1.$$

To verify the last equality without differentiating a singular logarithm at its pole, note that the integrand is $-\log|t|\,(d/dt)[t\phi(t-x)]$. Integrate by parts separately on $(-\infty,-\varepsilon)$ and (ε,∞). The remaining integral is $\int_{|t|>\varepsilon}\phi(t-x)\,dt$, and the boundary terms are $O(\varepsilon|\log\varepsilon|)$. Taking $\varepsilon\downarrow 0$ proves the identity. Solving $H' + xH = 1$ with $H(0) = 0$ yields the Dawson-type formula

$$H(x) = e^{-x^2/2}\int_0^x e^{t^2/2}\,dt. \tag{9}$$

In particular, $H(x) > 0$ for $x > 0$.

We next establish the strict comparison

$$H(x) < \frac{x}{x^2-2} < \frac{1}{\sqrt{x^2-4}}, \qquad x > 2. \tag{10}$$

Put $b(x) = x/(x^2-2)$ and $D = b - H$. A direct derivative calculation gives

$$D'(x) + xD(x) = \frac{x^2-6}{(x^2-2)^2} =: F(x). \tag{11}$$

At $x = \sqrt{6}$, equation (9) shows that $D(x) > 0$ is equivalent to

$$\frac{e^3}{4} - \int_0^1 e^{3u^2}\,du > 0. \tag{12}$$

Termwise integration of the uniformly convergent exponential series gives the left side as

$$\sum_{n=0}^{\infty}\frac{3^n}{n!}\left(\frac{1}{4}-\frac{1}{2n+1}\right).$$

The first five terms sum exactly to

$$-\frac{3}{4}-\frac{1}{4}+\frac{9}{40}+\frac{27}{56}+\frac{15}{32} = \frac{197}{1120} > 0. \tag{13}$$

Every later term is positive, proving (12). By (11), $(e^{x^2/2}D(x))' = e^{x^2/2}F(x)$ is negative on $(2,\sqrt{6})$ and positive on $(\sqrt{6},\infty)$. Integrating backward or forward from the positive anchor proves $D(x) > 0$ for all $x > 2$. The second inequality in (10) follows by squaring positive quantities and using

$$(x^2-2)^2 - x^2(x^2-4) = 4 > 0.$$

For $x(a) = \sqrt{a} + 1/\sqrt{a}$, scaling the logarithm gives

$$\begin{aligned} G(a) &= \tfrac{1}{2}\log a + J(x(a)), \\ G'(a) &= \frac{1}{2a} + \frac{a-1}{2a^{3/2}} H(x(a)). \end{aligned} \tag{14}$$

If $a \geq 1$, this derivative is positive by (9); at $a = 1$ it is exactly $1/2$. If $0 < a < 1$, then $x(a) > 2$ and

$$\frac{1}{\sqrt{x(a)^2 - 4}} = \frac{\sqrt{a}}{1-a}.$$

Using the strict upper bound (10) in the second line of (14), with its negative coefficient $a - 1$, again gives $G'(a) > 0$.

Finally, $G(a) \to 0$ as $a \downarrow 0$. Here are the required integrability details. Put $Y_a = 1 + a + \sqrt{a}\, Z$. On $|Y_a| \geq 1/2$, the negative part of $\log |Y_a|$ is bounded, while its positive part is uniformly integrable for $0 < a \leq 1$, since $\log^+ |Y_a| \leq |Y_a|$ and $\sup_{0<a\leq 1} \mathbb{E}|Y_a|^2 < \infty$. As $Y_a \to 1$ in probability, the expectation on this region tends to zero. On $|y| < 1/2$, the density of Y_a is bounded above by

$$(2\pi a)^{-1/2} \exp\left(-\frac{1}{8a}\right).$$

Multiplying this by the finite integral $\int_{-1/2}^{1/2} |\log |y||\, dy$ proves that the other contribution tends to zero. Strict monotonicity from this zero limit proves (7).

In dimension $d \geq 1$, the logarithmic norm moment is finite: bounded density and the radial integral of $r^{d-1}|\log r|$ control zero, and Gaussian tails control infinity. Pointwise, outside null sets,

$$\log \left\| (1+a)e + \sqrt{a}\, Z \right\|^2 \geq 2 \log |1 + a + \sqrt{a}\, Z_1|.$$

Taking expectations proves (8), including $d = 1$. ∎

4 Inverse moments and exit from small states

Lemma 2. *There exist $p \in (0, d/2)$, $\delta > 0$, and $q \in (0,1)$ such that*

$$\mathbb{E}[C_{n+1}^{-p} \mid C_n = c] \leq q c^{-p}, \qquad 0 < c \leq \delta. \tag{15}$$

Proof. For small $p \geq 0$ define $M(p, c) = \mathbb{E} W(c, Z)^{-p}$. At the boundary,

$$M(0,0) = 1, \qquad \partial_p^+ M(0,0) = -L < 0.$$

To justify the derivative, choose $0 < p_0 < d/2$ first. In the variable $y = (1+a)e + \sqrt{a}\, Z$, the Gaussian density is bounded near zero, and

$$\|y\|^{-2p_0} \big(1 + |\log \|y\|^2|\big)$$

is locally integrable there. Outside the unit ball, Gaussian tails control the logarithmic factors. These bounds dominate the difference quotients for $0 \le p \le p_0$, so dominated convergence applies. There is therefore a fixed $p \in (0, d/2)$ with $M(p, 0) < 1$.

For this p, express $M(p, c)$ as the integral of $\|y\|^{-2p}$ against the Gaussian density of mean $(1 + a - \eta c)e$ and covariance aI_d. For c in a bounded neighborhood of zero, these densities are uniformly bounded near zero and have a common integrable Gaussian envelope outside a sufficiently large ball. Dominated convergence proves continuity at $c = 0$. Choose $\delta > 0$ and $q \in (M(p, 0), 1)$ small enough that $M(p, c) \le q$ for $0 \le c \le \delta$. Multiplication by c^{-p} proves (15). ∎

Every transition from a strictly positive finite state is positive and finite almost surely, because a nondegenerate Gaussian vector is finite and assigns probability zero to each point. Induction and a countable union prove the finite-time assertion in Theorem 1.

Lemma 3. *For every deterministic $c > 0$, a chain started at $C_0 = c$ satisfies*

$$\mathbb{P}_c(C_n > \delta \text{ infinitely often}) = 1. \tag{16}$$

Proof. First take $0 < c \le \delta$, and let $T = \inf\{n \ge 0 : C_n > \delta\}$. We claim

$$\mathbb{E}_c[C_n^{-p} \mathbf{1}_{\{T > n\}}] \le q^n c^{-p}. \tag{17}$$

At $n = 0$ this is equality. On $T > n$, the current state lies in $(0, \delta]$. Conditioning on the past, dropping the additional killing at time $n + 1$ only enlarges the nonnegative expectation; then (15) applies. Induction proves (17) without an optional-stopping argument. On $T > n$, $C_n^{-p} \ge \delta^{-p}$, so

$$\mathbb{P}_c(T > n) \le \delta^p q^n c^{-p} \longrightarrow 0.$$

Thus the process cannot stay in $(0, \delta]$ forever from any positive state in that interval. At each deterministic time m, the Markov property now gives probability zero to $\{C_n \le \delta \text{ for every } n \ge m\}$; if $C_m > \delta$, this event is already impossible. A countable union over m proves (16). No positive lower bound on the realized states is assumed. ∎

5 Uniform positive escape probability

Choose $R > \delta$ large enough to satisfy all three inequalities

$$R \ge \frac{2(1 + a)}{\eta}, \qquad \frac{\eta^2 R^2}{16} \ge 2, \qquad \frac{64ad}{3\eta^2 R^2} \le \frac{1}{2}. \tag{18}$$

Such an R exists for every fixed set of parameters.

Lemma 4. *For $h(c) = \mathbb{P}_c(C_n \to +\infty)$, one has*

$$h(c) \ge \frac{1}{2} \qquad (c \ge R). \tag{19}$$

Proof. Let $r_j = 2^j R$ and consider the noise events

$$B_j = \left\{ \|Z_j\| \le \frac{\eta r_j}{4\sqrt{a}} \right\}, \qquad j \ge 0.$$

Starting from $c \ge R$, induction on j shows that on $B_0 \cap \cdots \cap B_j$ one has $C_j \ge r_j$ and

$$\begin{aligned} |1 + a - \eta C_j| &\ge \eta C_j/2, \\ \left\|(1 + a - \eta C_j)e + \sqrt{a}\, Z_j\right\| &\ge \eta C_j/4, \\ C_{j+1} &\ge \eta^2 C_j^3/16 \ge 2C_j \ge r_{j+1}. \end{aligned}$$

Indeed, $C_j \ge r_j$ makes the noise norm at most $\eta C_j/4$, while (18) gives the other bounds. Consequently $\bigcap_{j\ge 0} B_j$ implies $C_j \to +\infty$. Markov's inequality, using $\mathbb{E}\|Z_j\|^2 = d$, gives

$$\mathbb{P}(B_j^c) \le \frac{16ad}{\eta^2 R^2\, 4^j}.$$

The countable union bound therefore gives

$$h(c) \ge \mathbb{P}\left(\bigcap_{j\ge 0} B_j \right) \ge 1 - \frac{64ad}{3\eta^2 R^2} \ge \frac{1}{2}.$$

Independence of the events is available but unnecessary for this bound. ∎

Lemma 5. *There is $\epsilon > 0$ such that $h(c) \ge \epsilon$ for every $c > \delta$.*

Proof. Define the finite constants

$$B = \max_{\delta \le c \le R} |1 + a - \eta c|, \qquad K = \frac{B + \sqrt{R/\delta}}{\sqrt{a}}, \qquad \beta = \mathbb{P}(\|Z\| \ge K) > 0.$$

The Gaussian norm has a strictly positive tail at every finite threshold, including when $d = 1$. If $c \in [\delta, R]$ and $\|Z\| \ge K$, the reverse triangle inequality gives

$$\left\|(1 + a - \eta c)e + \sqrt{a}\, Z\right\| \ge \sqrt{R/\delta}, \qquad cW(c, Z) \ge R.$$

By the Markov property and Lemma 4, $h(c) \ge \beta/2$ on $[\delta, R]$, while $h(c) \ge 1/2$ on $[R, \infty)$. Thus $\epsilon = \beta/2$ works. ∎

6 From repeated chances to almost sure escape

Proof of Theorem 1. The finite-time positivity and finiteness have already been proved, and the zero-state assertion follows immediately from (1). Define

$$\mathcal{F}_n = \sigma(C_0, Z_0, \ldots, Z_{n-1}), \qquad A = \{C_n \to +\infty\}, \qquad \mathcal{F}_\infty = \sigma\left(\bigcup_n \mathcal{F}_n \right).$$

The event A belongs to $\mathcal{F}_\infty$, by its countable limit-event definition. The measurable recursion also makes h measurable. Deleting finitely many terms does not affect convergence to infinity. Independent future noises and the time-homogeneous transition rule therefore imply

$$\mathbb{E}[\mathbf{1}_A \mid \mathcal{F}_n] = h(C_n). \tag{20}$$

By upward conditional-expectation convergence, the bounded martingale on the left converges almost surely to $\mathbb{E}[\mathbf{1}_A \mid \mathcal{F}_\infty] = \mathbf{1}_A$.

Lemma 3 says that almost every path has infinitely many n for which $C_n > \delta$. At each of these times, Lemma 5 gives $h(C_n) \geq \epsilon > 0$. The almost-sure limit in (20) consequently cannot be zero. Since it equals the indicator $\mathbf{1}_A$, it must equal one almost surely. Hence $\mathbb{P}(A) = 1$. ∎

This argument proves full convergence to infinity, not merely an unbounded limsup. It does not use a tail zero-one law for a state-dependent chain, and it does not identify positive-probability escape with almost-sure escape.

7 Scope and reproducibility

The conclusion concerns the empirical covariance of the fixed-step, untamed update (3) for a one-dimensional standard Gaussian target. It makes no assertion about the centroid, individual particles or coordinates, an escape rate, variable step sizes, or higher-dimensional targets. The regularized and leave-one-out processes considered elsewhere in [1] have different updates and are not covered.

The exponential growth event in Lemma 4 is only a device for obtaining a uniform positive escape probability. Its growth bound is not asserted for almost every trajectory. The boundary comparison, the inverse-moment argument, and the martingale upgrade are all analytic; no numerical simulation or numerical positivity test enters the proof.

The accompanying source archive contains this manuscript, its style and build files, and a standard-library Python checker for the finite rational identities used in the proof. The checker verifies (13), the numerator identity in (10), the derivative identity behind (11), and the geometric-series coefficient $16(4/3) = 64/3$. It supplements, rather than replaces, the analytic arguments, including the positivity of the infinite series tail.

For precise source identification, the official version used is arXiv:2609.03326v1, accessed October 6, 2026. The relevant derivation is in Section 3.1, printed pages 7–8, equations (14)–(16). Its PDF SHA-256 is

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95a41005562f010b1ee7c40ee4b159b
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The two displayed lines concatenate to the full hash. The source archive provides the official links and this byte pin, rather than redistributing the paper.

References

[1] H. Y. Tan and Y. Chen. Geometric Ergodicity of Affine Invariant Ensemble Langevin and its Discrete Time Variants. arXiv:2609.03326v1, September 3, 2026. https://arxiv.org/abs/2609.03326v1.